

Lab 5.1: Examining Wireless Networks

Remove Lab Requirement Instructions in your submission

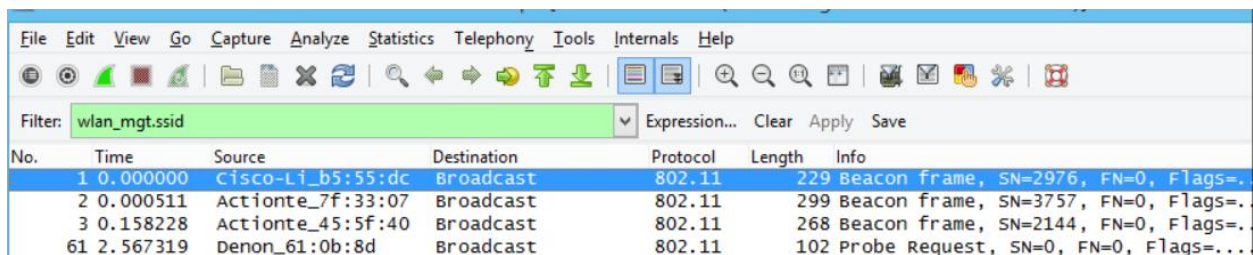
Lab Requirements:

1. In this assignment you will complete the assigned labs according to the instructions. This document will serve as a template to fill in answers where noted and screenshots required for each question.
2. Please make sure your screenshots follow the requirements for the course. You do not want to lose points needlessly for improper screenshots or formatting.
3. **Please make sure you show the syntax in your screenshots!** In order to get credit for your screenshots, you must show syntax and the output.
4. It is expected that any and all external sources used to complete any labs or support answers are properly cited in APA format and in a References Section at the end of the lab document.
5. Please name the file Lastname_LabTitle_Week# using your own last name at the beginning of the filename, lab title, and the appropriate week number. For example, if I were to name this lab: **Allen_Lab5.1_Week5.pdf**
6. Submit your work to the appropriate Assignment/Activity link in Blackboard. Do not submit assignments via course messages.
7. Points are assigned based on the completeness of your answers. Points will be deducted for errors and omissions.
8. There will be challenge flags throughout these labs. These are not required to be completed for the assignments, but feel free to complete them on your own. Also feel free

to go through the labs as many times as you want. Going through them a couple of times can help students understand the material.

Lab 5.1: Part 1 – Viewing Wireless Networks and Connected Devices

1. Question 4



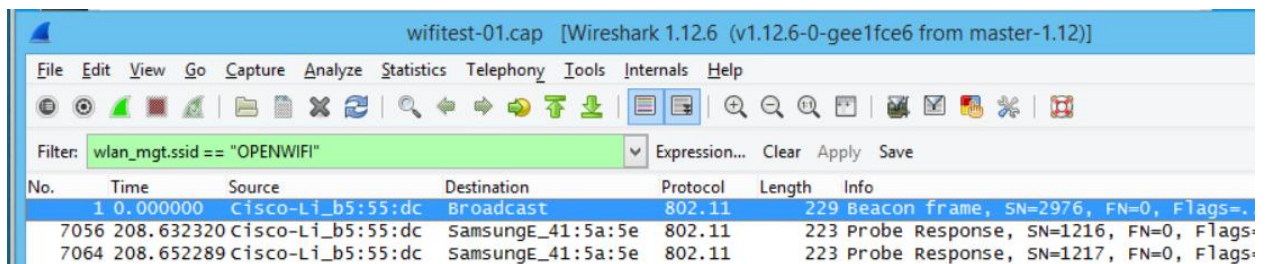
The image shows a Wireshark capture of wireless network traffic. The filter bar at the top is set to 'wlan_mgt.ssid'. The packet list shows four packets, all of which are 802.11 management frames (Beacon frames or Probe Request) with a length of 229 or 102 bytes. The source and destination addresses are broadcast (ff:ff:ff:ff:ff:ff).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco-Li_b5:55:dc	Broadcast	802.11	229	Beacon frame, SN=2976, FN=0, Flags=.
2	0.000511	Actionte_7f:33:07	Broadcast	802.11	299	Beacon frame, SN=3757, FN=0, Flags=.
3	0.158228	Actionte_45:5f:40	Broadcast	802.11	268	Beacon frame, SN=2144, FN=0, Flags=.
61	2.567319	Denon_61:0b:8d	Broadcast	802.11	102	Probe Request, SN=0, FN=0, Flags=...

2. What does this filter show you?

management frames that contain an SSID element

3. Question 5



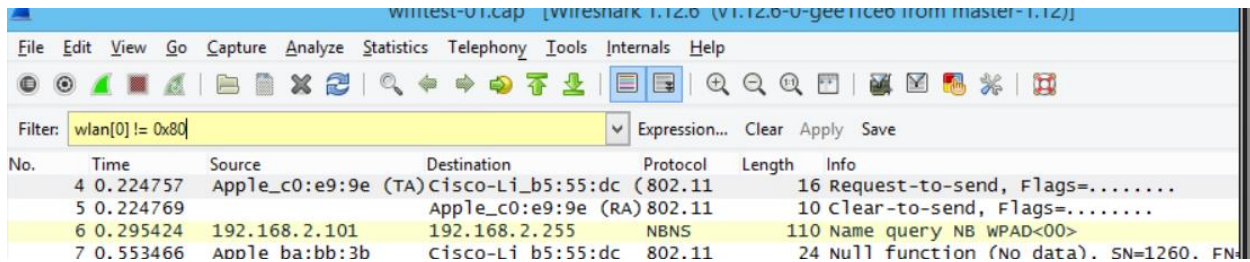
The image shows a Wireshark capture of wireless network traffic. The filter bar at the top is set to 'wlan_mgt.ssid == "OPENWIFI"'. The packet list shows four packets, all of which are 802.11 management frames (Beacon frames or Probe Response) with a length of 223 or 229 bytes. The source and destination addresses are broadcast (ff:ff:ff:ff:ff:ff).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco-Li_b5:55:dc	Broadcast	802.11	229	Beacon frame, SN=2976, FN=0, Flags=.
7056	208.632320	Cisco-Li_b5:55:dc	SamsungE_41:5a:5e	802.11	223	Probe Response, SN=1216, FN=0, Flags=.
7064	208.652289	Cisco-Li_b5:55:dc	SamsungE_41:5a:5e	802.11	223	Probe Response, SN=1217, FN=0, Flags=.

4. What does this filter show you?

management frames that carry the SSID "OPENWIFI"

5. Question 8



The screenshot shows the Wireshark interface with the filter `wlan[0] != 0x80` applied. The packet list shows four packets, with the third packet (No. 6) highlighted in yellow.

No.	Time	Source	Destination	Protocol	Length	Info
4	0.224757	Apple_c0:e9:9e (TA)	Cisco-Li_b5:55:dc (802.11)	802.11	16	Request-to-send, Flags=.....
5	0.224769	Apple_c0:e9:9e (RA)	Apple_c0:e9:9e (TA)	802.11	10	Clear-to-send, Flags=.....
6	0.295424	192.168.2.101	192.168.2.255	NBNS	110	Name query NB WPAD<00>
7	0.553466	Apple_ba:bb:3b	Cisco-Li_b5:55:dc	802.11	24	Null function (No data). SN=1260. FN=...

6. What does this filter show you?

It's excluding beacon frames.

`wlan[0] != 0x80` filters for 802.11 packets whose first byte of the Frame Control field isn't 0x80

7. Question 10

wifitest-01.cap [Wireshark 1.12.6 (v1.12.6-0-gee1fce6 from master-1.12)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: wlan.addr == 00:1C:B5:55:DC

Expression...

Clear

Apply

Save

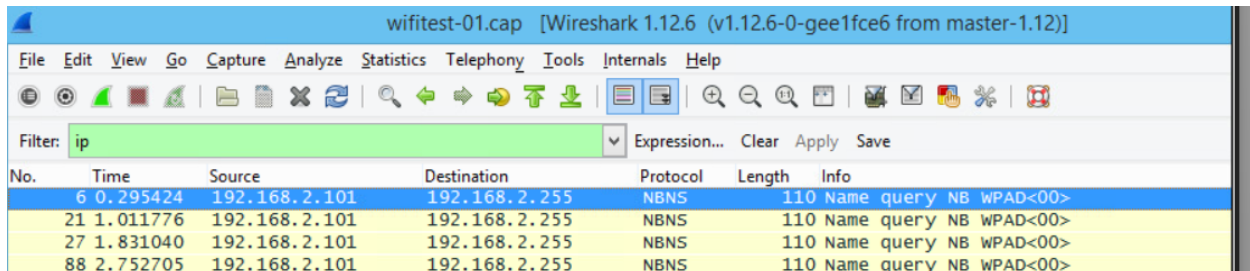
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco-Li_b5:55:dc	Broadcast	802.11	229	Beacon frame, SN=2976, FN=0, Flags=...
4	0.224757	Apple_c0:e9:9e (TA)	Cisco-Li_b5:55:dc (802.11)	802.11	16	Request-to-send, Flags=.....
6	0.295424	192.168.2.101	192.168.2.255	NBNS	110	Name query NB WPAD<00>
7	0.553466	Apple_ba:bb:3b	Cisco-Li_b5:55:dc	802.11	24	Null function (No data), SN=1260, FN=...
9	0.553466	Apple_ba:bb:3b (TA)	Cisco-Li_b5:55:dc (802.11)	802.11	16	Request-to-send, Flags=.....
11	0.570370	Cisco-Li_b5:55:dc (Apple_ba:bb:3b (RA)	802.11	16	Request-to-send, Flags=.....	
12	0.570362	Cisco-Li_b5:55:dc	Cisco-Li_b5:55:dc (802.11)	802.11	10	Clear-to-send, Flags=.....
13	0.570874	Apple_ba:bb:3b (TA)	Cisco-Li_b5:55:dc (802.11)	802.11	16	Request-to-send, Flags=.....

8. What does this filter show you?

It's a MAC address filters that shows all frames that involve the device

Lab 5.1: Part 2 – Viewing Wireless Traffic above Layer 2

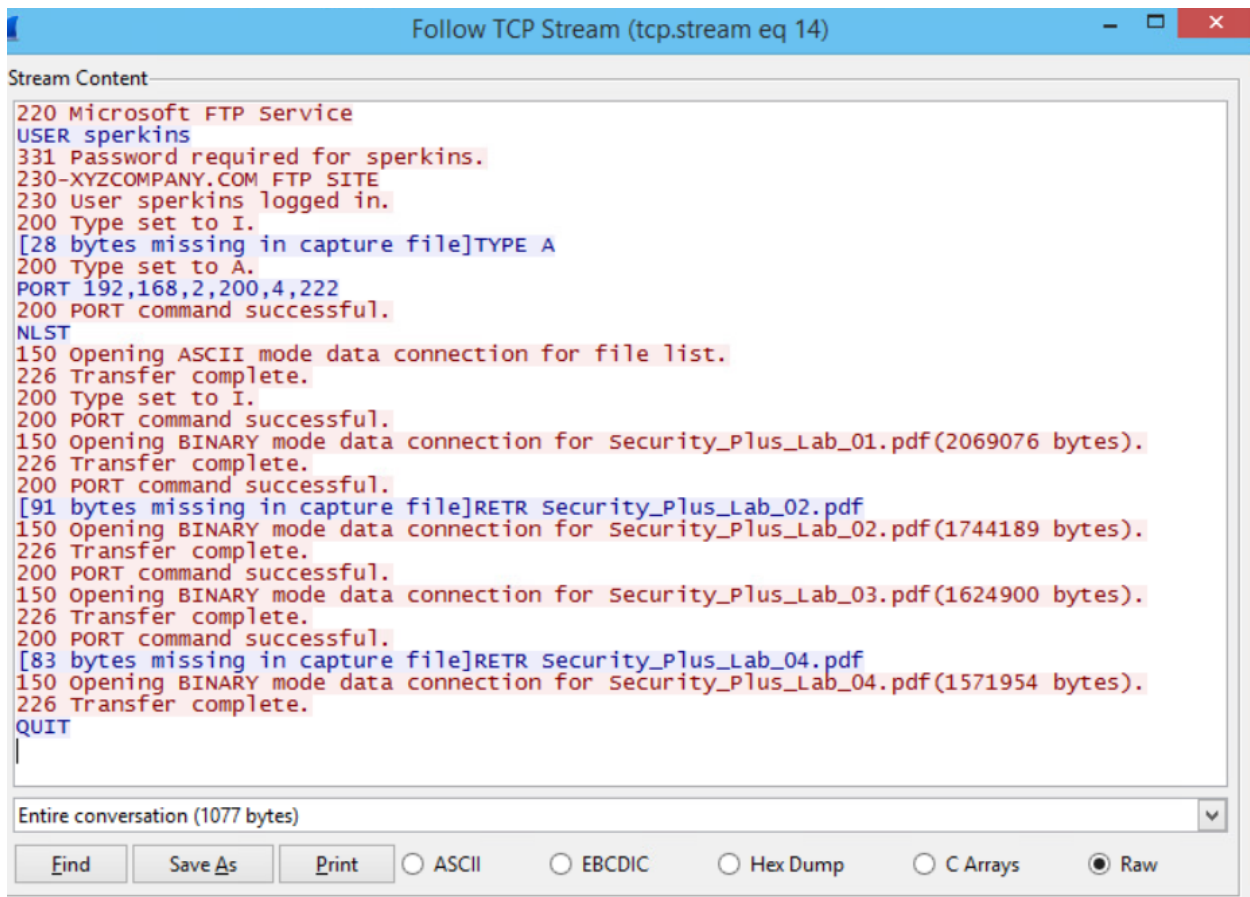
1. Question 1



The image shows a Wireshark packet capture window titled "wifitest-01.cap [Wireshark 1.12.6 (v1.12.6-0-gee1fce6 from master-1.12)]". The filter is set to "ip". The packet list shows four packets, all of which are NBNS Name queries from 192.168.2.101 to 192.168.2.255. The packet details pane shows the structure of an NBNS query.

No.	Time	Source	Destination	Protocol	Length	Info
6	0.295424	192.168.2.101	192.168.2.255	NBNS	110	Name query NB WPAD<00>
21	1.011776	192.168.2.101	192.168.2.255	NBNS	110	Name query NB WPAD<00>
27	1.831040	192.168.2.101	192.168.2.255	NBNS	110	Name query NB WPAD<00>
88	2.752705	192.168.2.101	192.168.2.255	NBNS	110	Name query NB WPAD<00>

2. Question 4



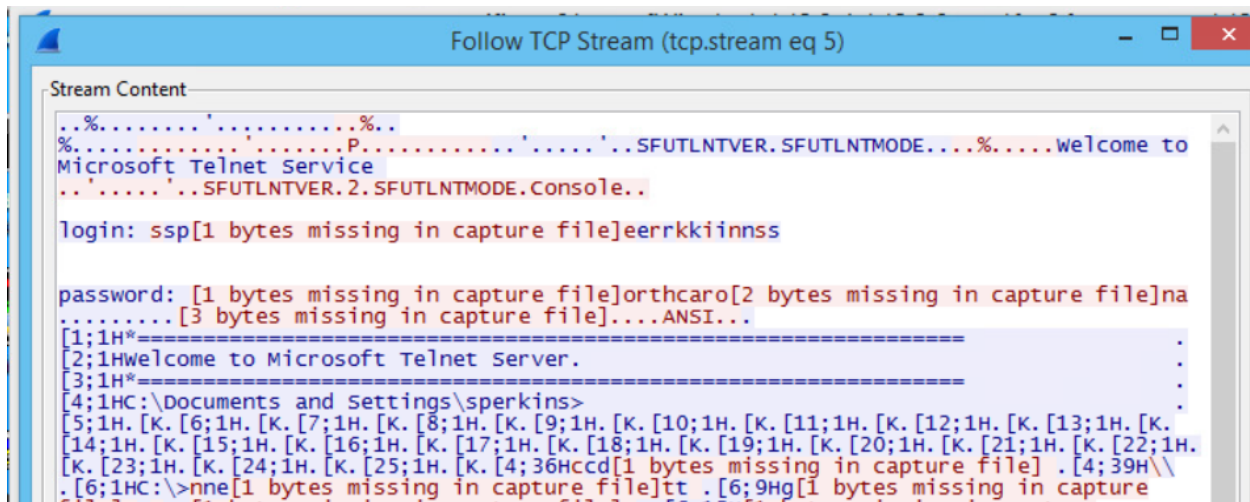
3. Question 7



```
Follow TCP Stream (tcp.stream eq 3)

Stream Content
+OK Microsoft Exchange Server 2003 POP3 server version 6.5.6944.0
(server.XYZCOMPANY.COM) ready.
USER sperkins
+OK
PASS northcarolina
+OK User successfully logged on.
STAT
+OK 1 1852
+OK
1 AAgSjAAAAgVYyr664+lmwOPwhfiQMxm
.
+OK 1 1852
1 1852
.
[12 bytes missing in capture file]RETR 1
+OK
Received: from studentPC ([192.168.2.200]) by server.XYZCOMPANY.COM with Microsoft
SMTPSVC(6.0.3790.0);
. Sun, 8 May 2016 13:41:42 -0400
```

4. Question 10



```
Follow TCP Stream (tcp.stream eq 5)

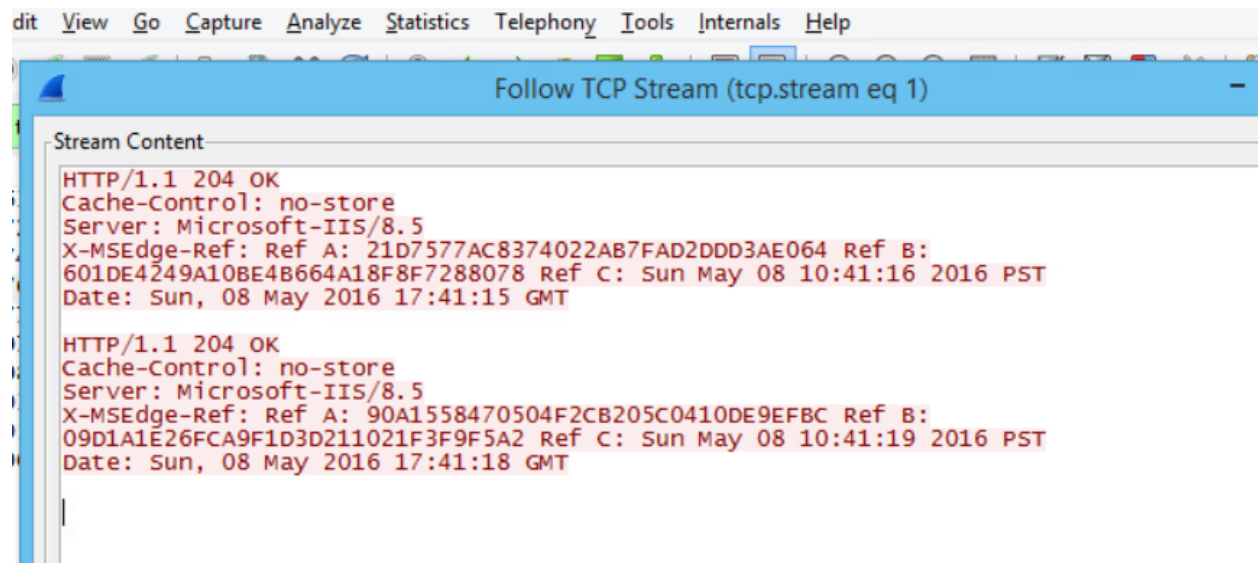
Stream Content
..%.....'.....%..
%.....P.....'.....'..SFUTLNTVER.SFUTLNTMODE....%.....Welcome to
Microsoft Telnet Service
..'.....'..SFUTLNTVER.2.SFUTLNTMODE.Console..

login: ssp[1 bytes missing in capture file]eerrkkiinnss

password: [1 bytes missing in capture file]orthcaro[2 bytes missing in capture file]na
.....[3 bytes missing in capture file]....ANSI...

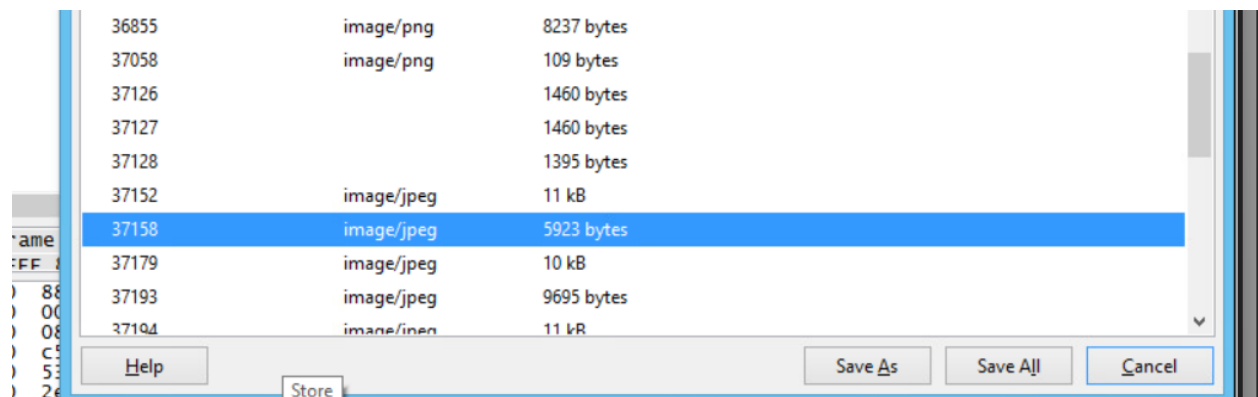
[1;1H*=====
[2;1HWelcome to Microsoft Telnet Server.
[3;1H*=====
[4;1HC:\Documents and Settings\sperkins>
[5;1H. [K. [6;1H. [K. [7;1H. [K. [8;1H. [K. [9;1H. [K. [10;1H. [K. [11;1H. [K. [12;1H. [K. [13;1H. [K.
[14;1H. [K. [15;1H. [K. [16;1H. [K. [17;1H. [K. [18;1H. [K. [19;1H. [K. [20;1H. [K. [21;1H. [K. [22;1H.
[K. [23;1H. [K. [24;1H. [K. [25;1H. [K. [4;36Hccd[1 bytes missing in capture file] .[4;39H\\
[6;1HC:>nne[1 bytes missing in capture file]tt .[6;9Hg[1 bytes missing in capture
file].....[1 bytes missing in capture file].....[6;15H[1 bytes missing in capture
```

5. Question 13

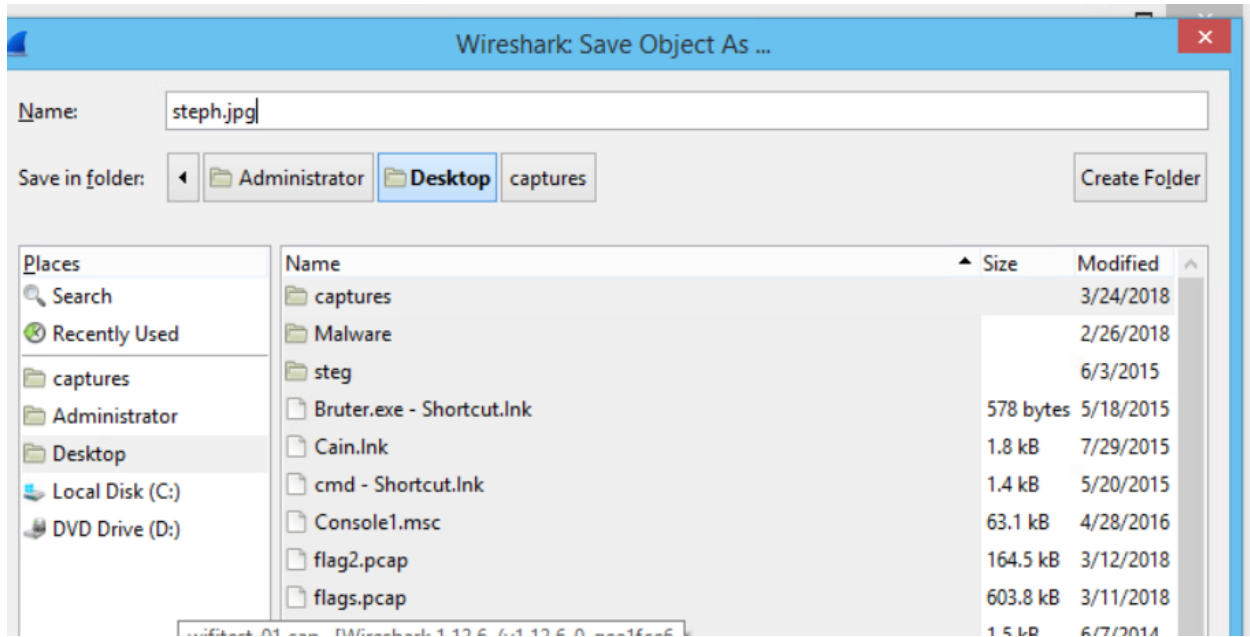


Lab 5.1: Part 3 – Parsing Object from Traffic

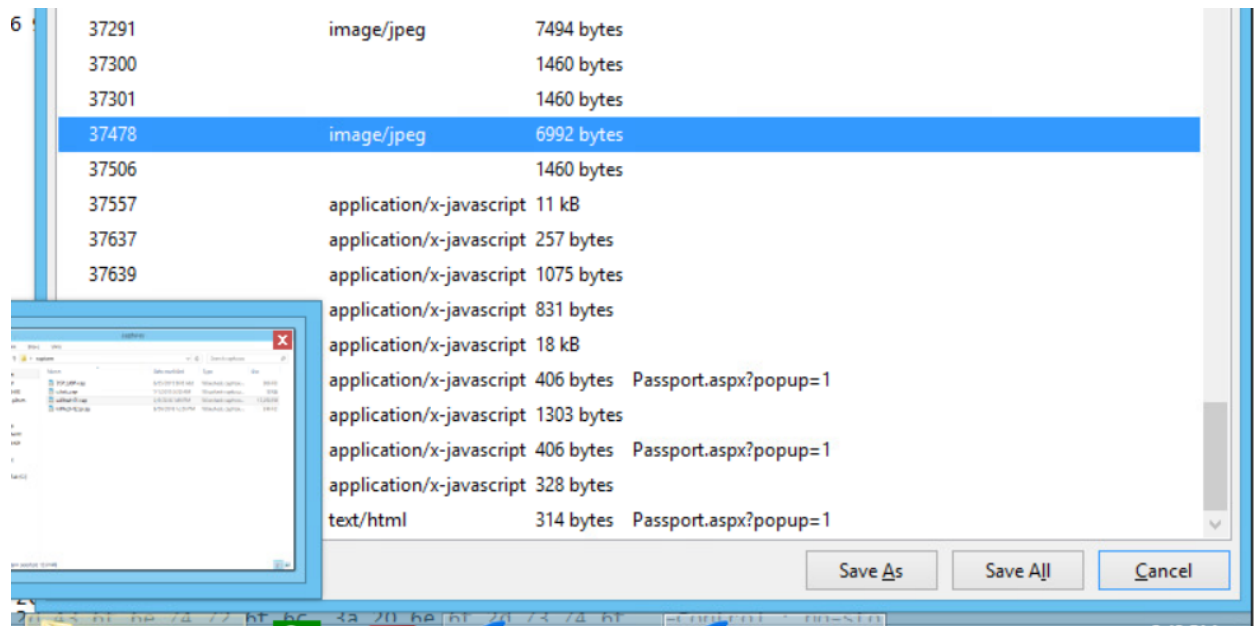
1. Question 2



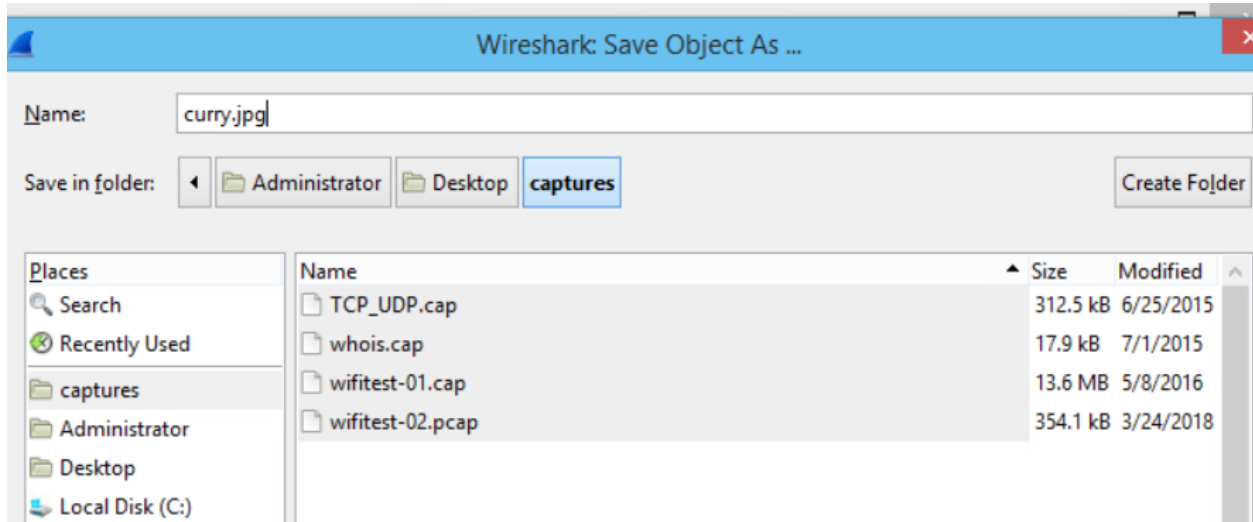
2. Question 3



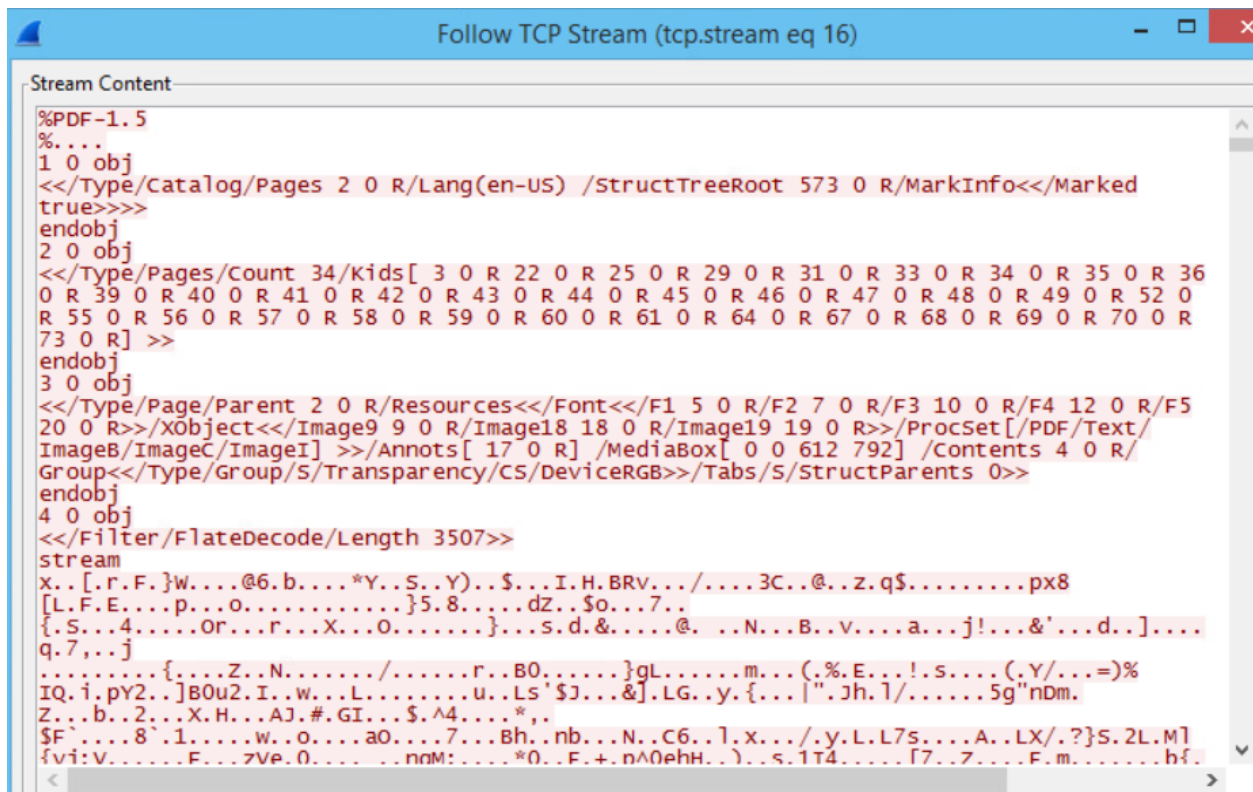
3. Question 4



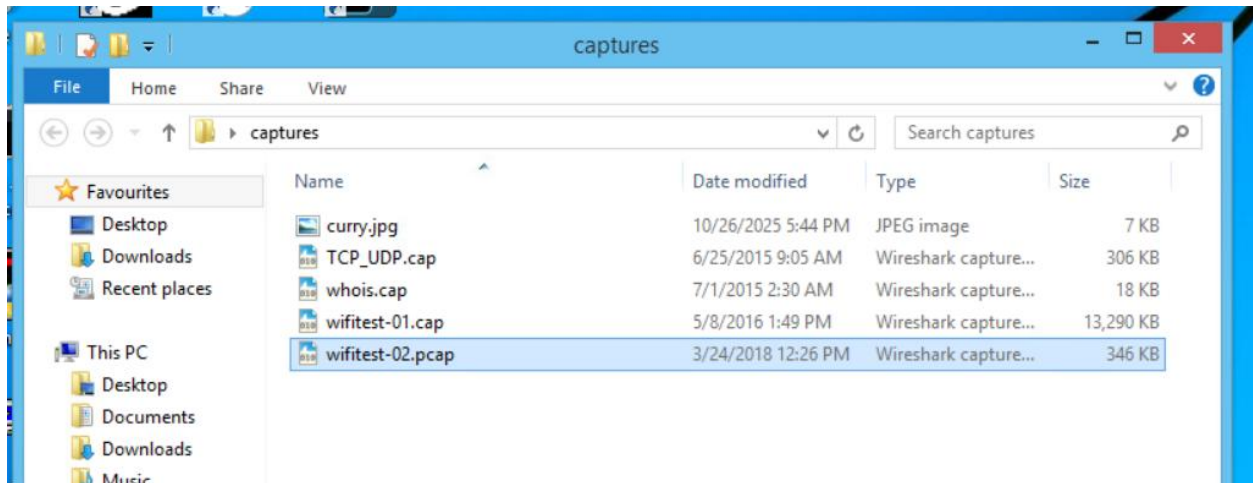
4. Question 5



5. Question 9



6. Question 17



Lab 5.1: Part 4 – Lessons Learned

This lab sharpened my ability to read and filter 802.11 management traffic with intent instead of just “watching packets go by.” Using `wlan_mgt.ssid` to surface SSID-bearing frames clarified how beacon, probe request/response exchanges actually advertise and discover networks. Filtering by a specific SSID (e.g., `wlan_mgt.ssid == "OPENWIFI"`) showed how quickly client/AP interactions can be isolated, while MAC-centric filters like `wlan.addr == <MAC>` tied activity back to a single device across roles (source/destination/BSSID). I also internalized how simple byte tests (e.g., excluding beacons via Frame Control checks) can declutter traces and make patterns visible. I was surprised by how much non-user traffic fills the air beacons, probes, and RTS/CTS before any higher-layer data appears. Another surprise was how easily a single mis-specified filter can hide critical frames; precision matters.